8. (a) Keeping warm / 88 W — 1A — 1

(b)

$$R_1 = \frac{V^2}{P} = \frac{220^2}{88}$$ 1M

$$= 550\ \Omega$$ 1A — 2

(c)

Total current $I_0 = \frac{P_0}{V} = \frac{550}{220} = 2.5\text{ A}$ — 1M

Current in R_1, $I_1 = \frac{220}{550} = 0.4\text{ A}$ — 1M

Current in R_2, $I_2 = 2.5 - 0.4 = 2.1\text{ A}$ — 1A — 3

Or Power to R_2, $550\text{ W} - 88\text{ W} = 462\text{ W}$

Current in R_2, $I_2 = \frac{P_2}{V} = \frac{462}{220}$

$= 2.1\text{ A}$

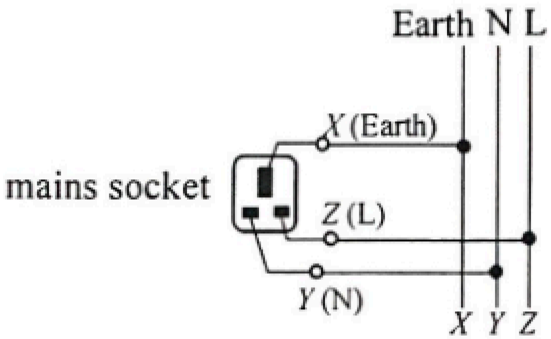
(d)

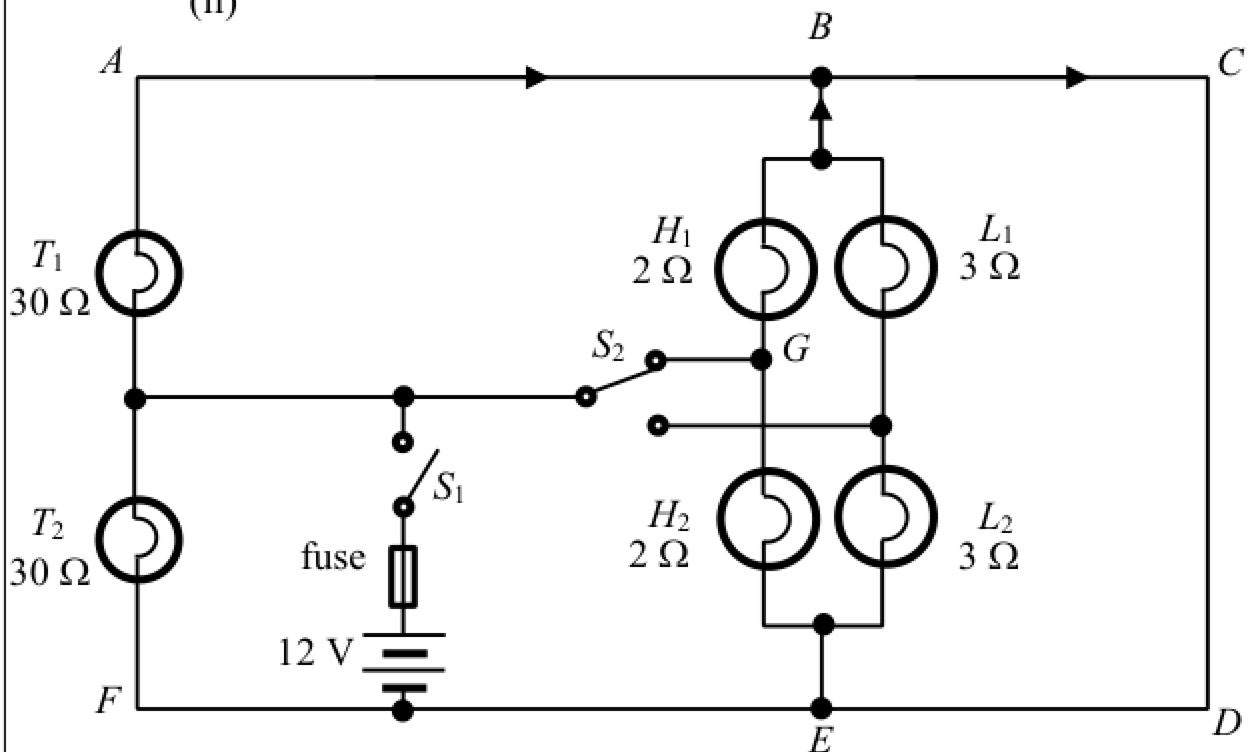
Peak current $= \sqrt{2}\,(2.5\text{ A})$ — 1M

$= 3.54\text{ A}$ — 1A — 2

8. (a) $P = \dfrac{V^2}{R}$ $500 = \dfrac{220^2}{R}$ $R = 96.8\ \Omega$	1A	Accept 97 Ω
	1	
(b) As the total resistance of the circuit for mode X is doubled,	1M	
total power dissipation $= \dfrac{V^2}{2R}$ $= \dfrac{220^2}{2 \times 96.8} = 250$ W	1A	Accept 249 W for 97 Ω
Alternative method: As the voltage across each heating element (1 and 2) is halved, power dissipated in each of the heating element P_1 or $P_2 = \dfrac{500}{4} = 125$ W (since $P \propto V^2$)	1M	
<u>Or</u> P_1 or $P_2 = \dfrac{V'^2}{R} = \dfrac{110^2}{96.8} = 125$ W	1M	
<u>Or</u> $i = \dfrac{V}{R_1 + R_2} = \dfrac{220}{2 \times 96.8} = 1.14$ A P_1 or $P_2 = i^2R = 1.14^2 \times 96.8 = 125$ W	1M	
Total power dissipation = 2 × 125 W = 250 W	1A	
	2	
(c) In mode Z, the equivalent resistance of the heating elements is the least as they are connected in parallel,	1A	
hence, under the same voltage, the total power dissipation is the largest since $P = \dfrac{V^2}{R}$.	1A	
	2	
(d) (i) For mode Z, the total power dissipated = 500 + 500 = 1000 W $I_z = \dfrac{P}{V} = \dfrac{1000}{220} = 4.55$ A	1M+1M	
Alternative method: $R_{eq} = \dfrac{96.8}{2}\ \Omega = 48.4\ \Omega$ $I = \dfrac{220\ \text{V}}{48.4\ \Omega} = 4.55$ A	1M+1M	
Most suitable value of fuse = 5 A	1A	
	3	
(ii) Although the heater still works in either connection, it is dangerous for switch S to be fitted in wire B (neutral) as the heater / cable would still be live even when the switch was turned off.	1A 1A	1+1
	2	
(iii) Wire C (Earth).	1A	
Current would be conducted from the case through this wire to the Earth.	1A	
	2	

	Marks	Remarks
8. (a) (i) To terminals X	1A	
	1	
(ii) $P = IV$		
$800\text{ W} = I(220\text{ V})$	1M	
$I = 3.636364\text{ A}$		
$\approx 3.64\text{ A}$	1A	
	2	
(iii) $800 = \dfrac{V^2}{R} + \dfrac{V^2}{4R} = \dfrac{5V^2}{4R}$	1M	
$P_{\text{keep warm}} = \dfrac{V^2}{4R}$	1M	
$= 800(\dfrac{1}{5}) = 160\text{ W}$	1A	
Alternative (I)		
$800 = \dfrac{V^2}{R} + \dfrac{V^2}{4R} = \dfrac{5V^2}{4R}$		
$R = 75.625\ \Omega$	1M	
$P_{\text{keep warm}} = \dfrac{V^2}{4R}$		
$= \dfrac{220^2}{4(75.625)}$	1M	
$= 160\text{ W}$	1A	
Alternative (II)		
Power $\propto \dfrac{1}{\text{resistance}}$	1M	
$\dfrac{P_{\text{keep warm}}}{P_{\text{heating}}} = \dfrac{R_{\text{eq}}}{4R}$		
$P_{\text{keep warm}} = P_{\text{heating}}\left(\left(\dfrac{1}{R} + \dfrac{1}{4R}\right)^{-1} / 4R\right)$	1M	
$= 160\text{ W}$	1A	
	3	
(b) (i) Electrical energy (consumed)	1A	
	1	
(ii) (1) only the fuse blows	1A	
(2) only the RCCB cuts off the supply	1A	
	2	

8. (a) 	1A	
	1	
(b) (i) - If one of the lighting sets / circuits fails, the other (in parallel) can still operate, i.e. both work independently. - Both can work at the rated power. - Any reasonable answer Any **ONE**	1A	
	1	
(ii) $P = IV$ $(300 + 450) = I(220)$ $\boxed{I = \frac{P_1}{V} + \frac{P_2}{V} = \frac{300}{220} + \frac{450}{220}}$	1M	
$I = 3.409091\text{ A} \approx 3.41\text{ A}$	1A	
Thus 5 A fuse should be used.	1A	
	3	
(c) Electrical energy used per day $= 0.500\text{ kW} \times 8\text{ h} + 2\text{ kW} \times 0.5\text{ h} + 3\text{ kW} \times 2\text{ h}$	1M	
$= 11\text{ kW h}$		
Cost $= \$0.9 / \text{kW h} \times 11\text{ kW h}$	1M	
$= \$9.9$	1A	
	3	

8. (a) Because L_1 and L_2 (connected in series) are <u>shorted by *BCDE*</u>	1A	Accept p.d. across each low-beam headlight (L_1 or L_2) is 0.
	1	
(b) (i) p.d. across T_2 = 12 V	1A	
	1	
(ii) 	1A 1A	Any two currents correctly marked. All currents correct.
largest current in branch *BC*	1A	
	3	
(c) Power supplied $P = 2 \times \left(\frac{12^2}{30} + \frac{12^2}{2}\right)$	1M	1M for same voltage across each bulb (T_1, T_2, H_1 and H_2) e.c.f. from (b)(i)
= 153.6 W = 154 W (3 sig. fig.)	1A	
$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6$ (or $\frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2}$)	1M	1M for all bulbs (T_1, T_2, H_1 and H_2) in parallel
$R_{eq} = 0.9375\ \Omega \approx 0.938\ \Omega \approx 1\ \Omega$ (3 sig. fig.)	1A	
	4	
(d) max. current $I_{max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8$ A (or $I_{max} = \frac{P}{V} = \frac{153.6}{12} = 12.8$ A)	1M	e.c.f. from (c)
Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.	1A	1A for 12.8 A and correct conclusion
	2	